

Refinement Changelog – FloodWay (Final vs. Preliminary)

Period: Preliminary Submission → Final Submission

Key Driver: Mentor (CEO) feedback, commercialization strategy, and real-world deployment readiness

Refinement Area	Preliminary System	Final System Refinement	Strategic Impact
Data Lifecycle & Historical Retention	Reports and uploaded images were removed after resolution.	Reports are permanently retained within the system and restricted from public view once resolved. Historical datasets are accessible to moderators and administrators for analytics and auditing purposes.	Enables long-term disaster trend analysis, AI retraining, institutional auditing, and data-driven policy evaluation.
Location Input Mechanism	Relied primarily on live GPS tracking during report submission.	Introduced manual map pinning alongside optional live GPS functionality. Report coordinates are stored as persistent metadata.	Improves reporting accuracy, supports pre-incident reporting, and increases usability in low-GPS or inaccessible environments.
IoT Sensor Integration	No physical sensing infrastructure integrated into the platform.	Integrated IoT water-level sensors into the reporting ecosystem, allowing sensor-based events to appear alongside community reports.	Establishes a hybrid sensing environment combining real-time physical monitoring with crowdsourced intelligence.
Incident Similarity & Data Structuring	Reports were treated as isolated incidents without relationship analysis.	Implemented similarity detection and grouped related reports into unified incident threads. Adopted structured smart-city compatible data standards including MQTT and JSON schema architecture.	Reduces duplicate reporting noise, improves situational awareness, and enhances interoperability with enterprise and municipal systems.

Platform Scope & Strategic Positioning	Focused exclusively on flood preparedness and evacuation.	Expanded into a scalable multi-disaster discovery platform with future support for landslides, haze, earthquakes, and generalized disaster reporting workflows.	Increases scalability, commercial relevance, and long-term market viability.
Monetization Strategy	No defined business or revenue model.	Introduced API-based commercialization strategy targeting logistics providers, government agencies, NGOs, and research institutions through tiered service access.	Establishes a sustainable business model centered on verified real-time disaster intelligence.
Simulation System	Generic 3D flood visualization with demonstration-oriented objectives.	Redesigned simulation environment to enhance danger visualization.	Transforms simulation from conceptual visualization into a practical disaster preparedness and emergency training asset.
UI/UX Prioritization	Balanced navigation structure with equal emphasis across features.	Reorganized interface using danger-first logic prioritizing active threats, verified reports, safe routing, and shelter guidance before secondary features.	Aligns interface behavior with real-world emergency response requirements and improves decision-making efficiency during crises.
Core Value Proposition	Emphasized system functionality and application features.	Repositioned the platform around verified disaster intelligence, real-time reporting reliability, and API-driven accessibility.	Strengthens investor positioning, commercialization potential, and enterprise adoption readiness.